

Setting up experiments

EXPERIMENTAL DESIGN IN PYTHON



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Experimental Design definition

- A process
- An objective and controlled way
- Draw specific conclusions
 - In reference to a hypothesis



¹ <https://www.sciencedirect.com/topics/earth-and-planetary-sciences/experimental-design>

Forming robust statements

X **probably** had an effect on Y. There is **likely** some **small** risk of error

- **Precise and Quantified language**

P-value analysis indicates X **had an effect on Y** with a **10% risk of Type I error**

- **Type I error**: incorrectly reject null hypothesis
- **Goal: Experimental design and statistical analysis**

Why experimental design?

Useful in many fields:

- Medical research
- Marketing
- Product analytics
- Agriculture
- Government policy



Some terminology...

- **Subjects** = *what* we are experimenting on (people, employees, users, etc.)



= Subject

Some terminology...

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- **Treatment** = some change given to one group



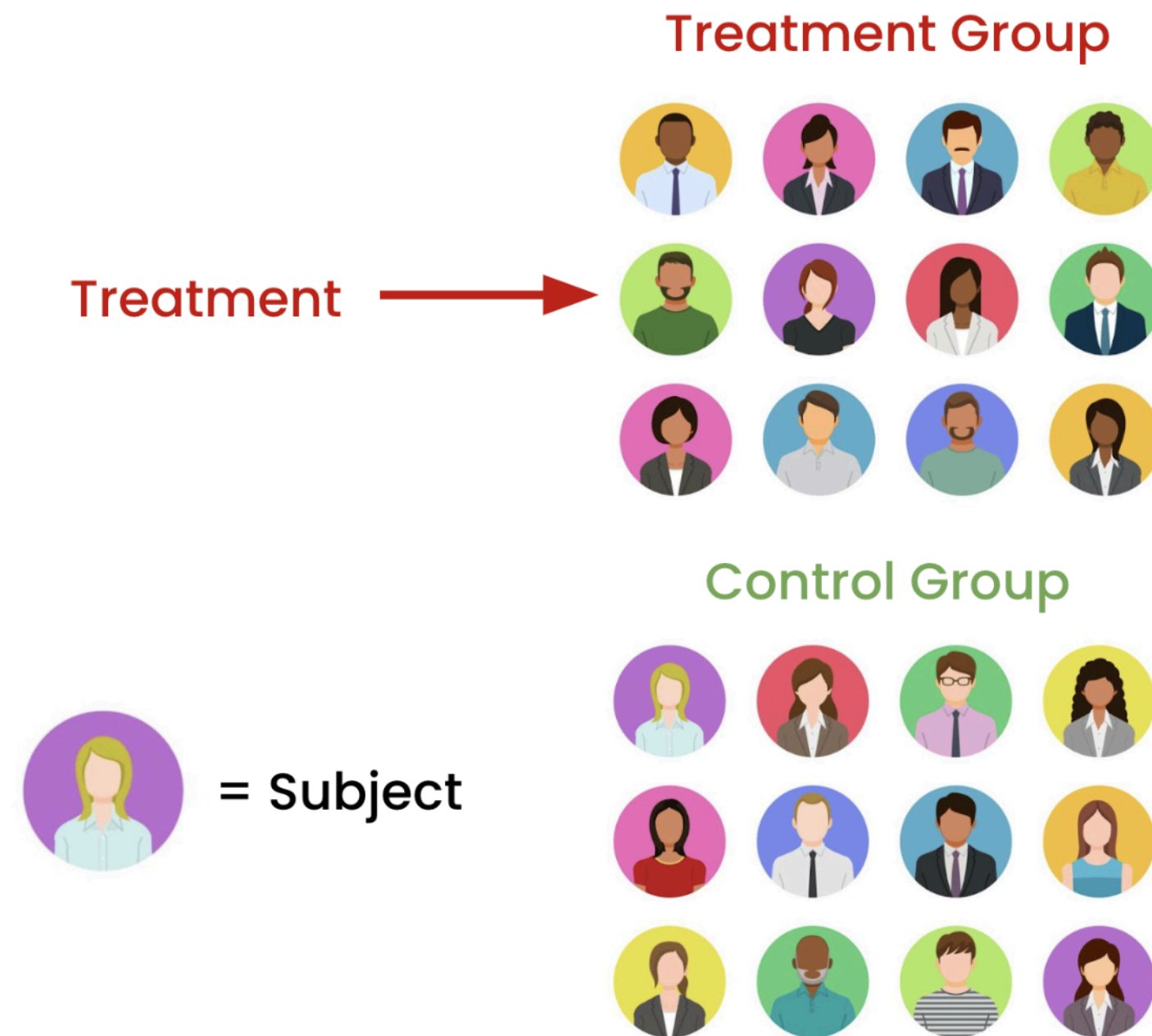
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Some terminology...

- **Subjects** = *what* we are experimenting on (people, employees, users, etc.)
- **Treatment** = some change given to one group
- **Control** = the group not given any change



Assigning subjects to groups

- How to assign subjects to groups?
 - Option 1 - *non-random* ('split' the DataFrame)
 - Option 2 - *random assignment*

- **Example:** 200 subjects heights in `heights` DataFrame

```
   id  height
0    0  177.98
1    1  174.17
2    2  178.89
```

Non-random assignment

Assignment by slicing the DataFrame

```
group1_nonrandom = heights.iloc[0:100,:]
group2_nonrandom = heights.iloc[100:,:]
compare_df = pd.concat(
    [group1_nonrandom['height'].describe(),
     group2_nonrandom['height'].describe()],
    axis=1)
compare_df.columns = ['group1', 'group2']
print(compare_df)
```

- Very different! (Mean 9cm apart)

	group1	group2
count	100.00	100.00
mean	170.32	179.19 <--
std	3.28	3.50
min	159.28	175.03
25%	168.06	176.57
50%	170.75	178.03
75%	173.09	180.79
max	174.92	191.32

Random assignment

- Use `.sample()`
 - `n` or `frac` (fraction 0-1)

```
group1 = heights.sample(frac=0.5,  
                        replace=False,  
                        random_state=42)  
group2 = heights.drop(group1.index)
```

```
print(compare_df)
```

- Much closer! (<1cm)

	group1	group2
count	100.00	100.00
mean	175.10	174.41 <--
std	5.39	5.78
min	163.07	159.28
25%	171.32	170.17
50%	175.22	174.86
75%	178.32	177.85
max	189.78	191.32

Assignment summary

- Subjects should be *randomly* assigned to groups
 - Observed changes correctly attributed
- Random subject assignment: `.sample()`
- Verify differences: `.describe()`

Treatment Group



Control Group

Let's practice!

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Experimental data setup

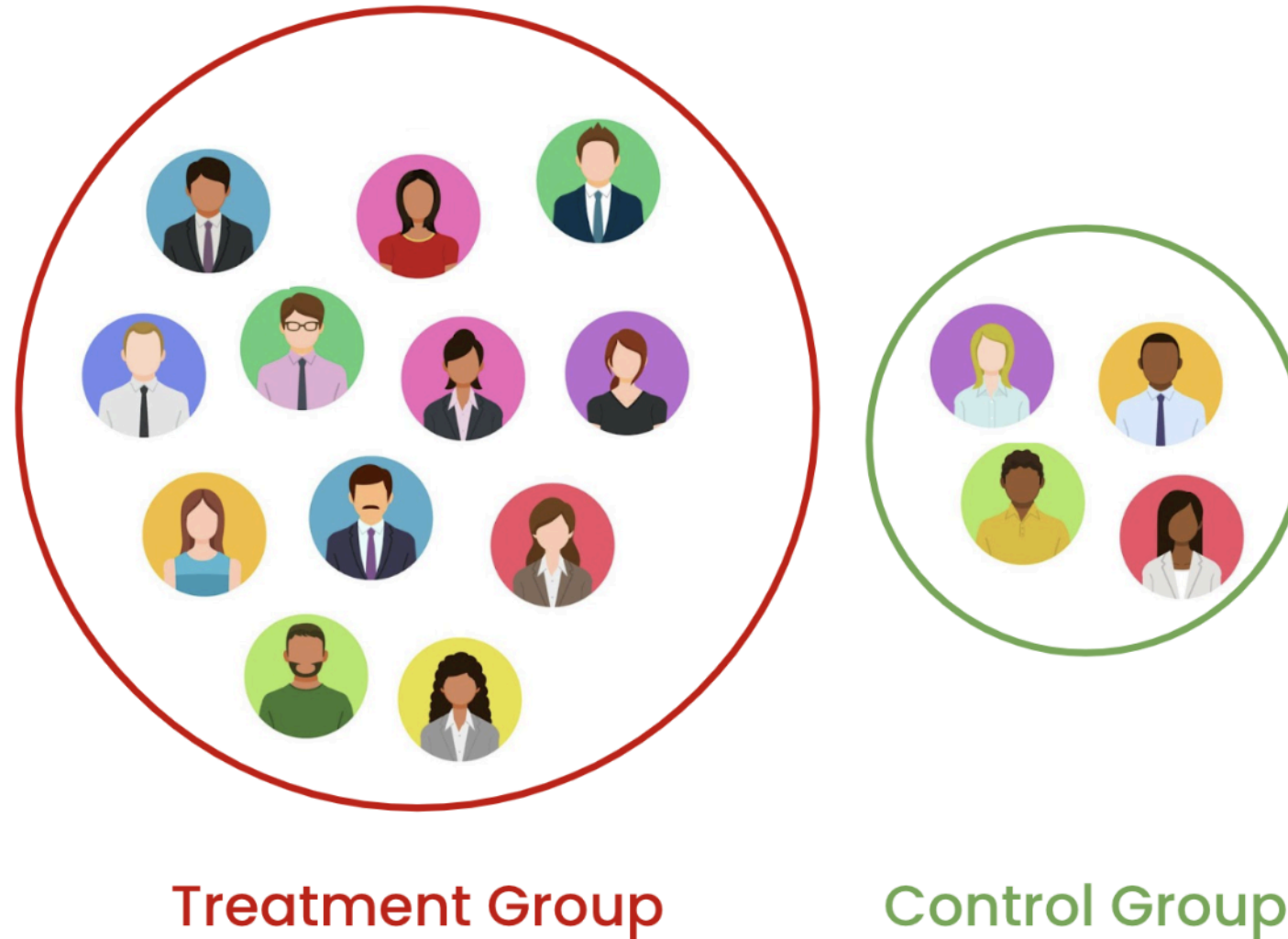
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The problem with randomization

1) Uneven issue: different number of subjects in groups



The problem with randomization

2) Covariate issue: High variability in some covariates → group imbalances in randomization

Does age impact reaction times?

Control Group



Physically Active

Treatment Group



Not Physically Active

Physical Activity =
Covariate

Result: harder to measure treatment effect!

Block randomization

- Split into blocks of size n **first** then randomly split
 - Fixes uneven issue



- 24 subjects split into two groups then randomized

Our dataset

- E-commerce dataset (ecom) (1000 subjects)
 - Average basket size
 - Average time on the website
 - Power user (Average 40+ daily minutes on website)

```
basket_size  web_time  power_user
0           227         7         0
1           123         5         0
2            98        16         0
3           211        45         1
4           133        17         0
```

Block randomization in Python

```
group1 = ecom.sample(frac=0.5, random_state=42, replace=False)
group1['Block'] = 1
group2 = ecom.drop(group1.index)
group2['Block'] = 2
print(len(group1), len(group2))
```

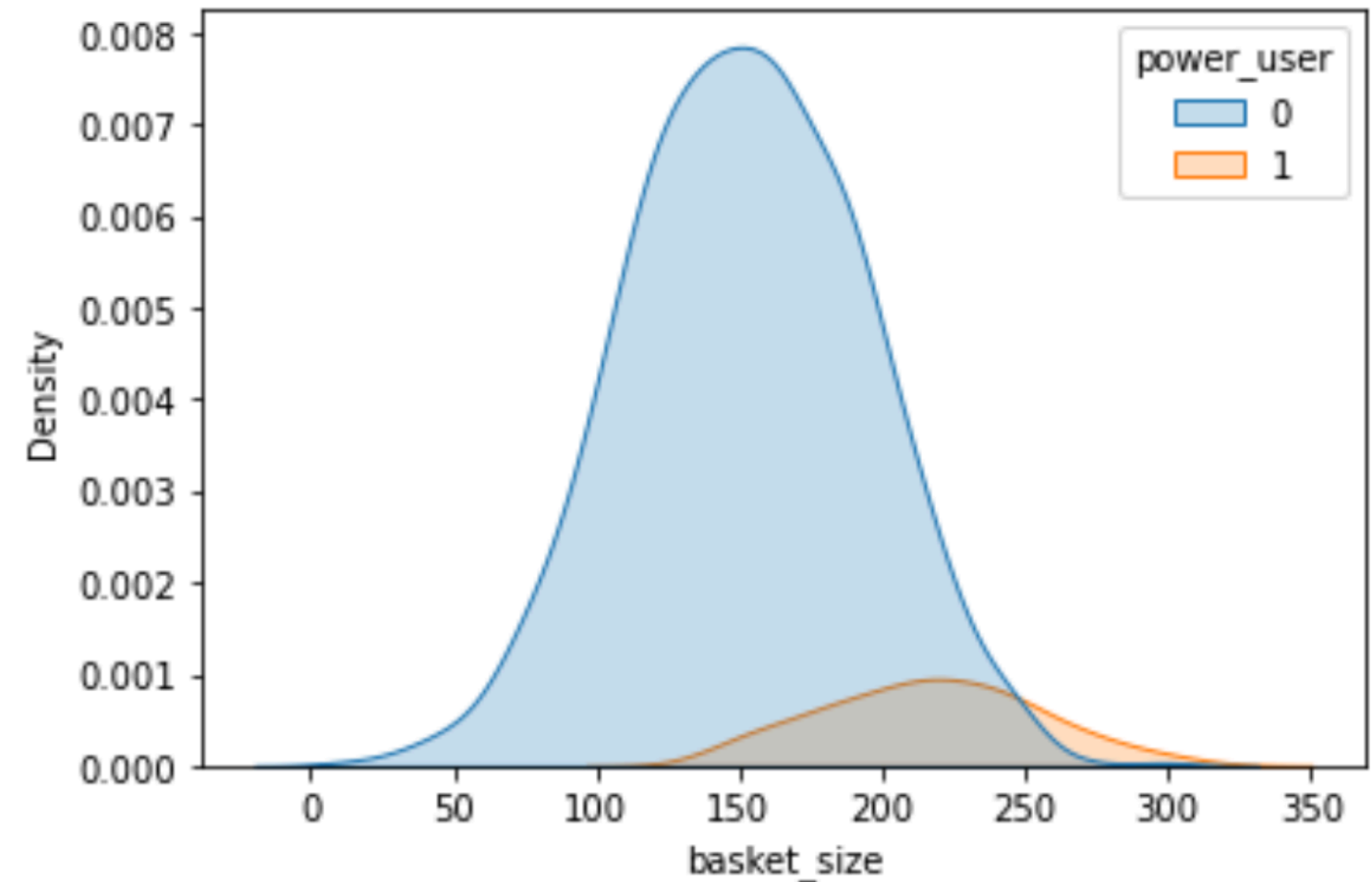
500,500

Visualizing splits

```
import seaborn as sns
import matplotlib.pyplot as plt
sns.displot(data=ecom,
            x='basket_size',
            hue='power_user',
            fill=True,
            kind='kde')

plt.show()
```

Confounding = variable *might* cause the effect rather than treatment



Stratified randomization

- Splitting based on covariate(s) *first*
 - Then randomization
- Green = All power users (Yellow = Not power users)
 - Then split Treatment/Control
- Fixes covariate/confounding issue
- Can be done for multiple covariates - but gets complex!

Not Power User

T	T
C	T
C	T
T	T
C	C
C	C

T	C
C	T
T	C
T	C

Power Users

Our first strata

- Separate power users
- Sample into Treatment/Control

```
strata_1 = ecom[ecom['power_user'] == 1]
strata_1['Block'] = 1
strata_1_g1 = strata_1.sample(frac=0.5, replace=False)
strata_1_g1['T_C'] = 'T'
strata_1_g2 = strata_1.drop(strata_1_g1.index)
strata_1_g2['T_C'] = 'C'
```

The second strata

- Separate not power users
- Sample into Treatment/Control

```
strata_2 = ecom.drop(strata_1.index)
strata_2['Block'] = 2
strata_2_g1 = strata_2.sample(frac=0.5, replace=False)
strata_2_g1['T_C'] = 'T'
strata_2_g2 = strata_2.drop(strata_2_g1.index)
strata_2_g2['T_C'] = 'C'
```

Confirming stratification

- Join blocks and groups
- Use groupby to check allocation

```
ecom_stratified = pd.concat([strata_1_g1, strata_1_g2, strata_2_g1, strata_2_g2])  
ecom_stratified.groupby(['Block', 'T_C', 'power_user']).size()
```

Block	T_C	power_user	
1	C	1	50
	T	1	50
2	C	0	450
	T	0	450

Let's practice!

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Normal data

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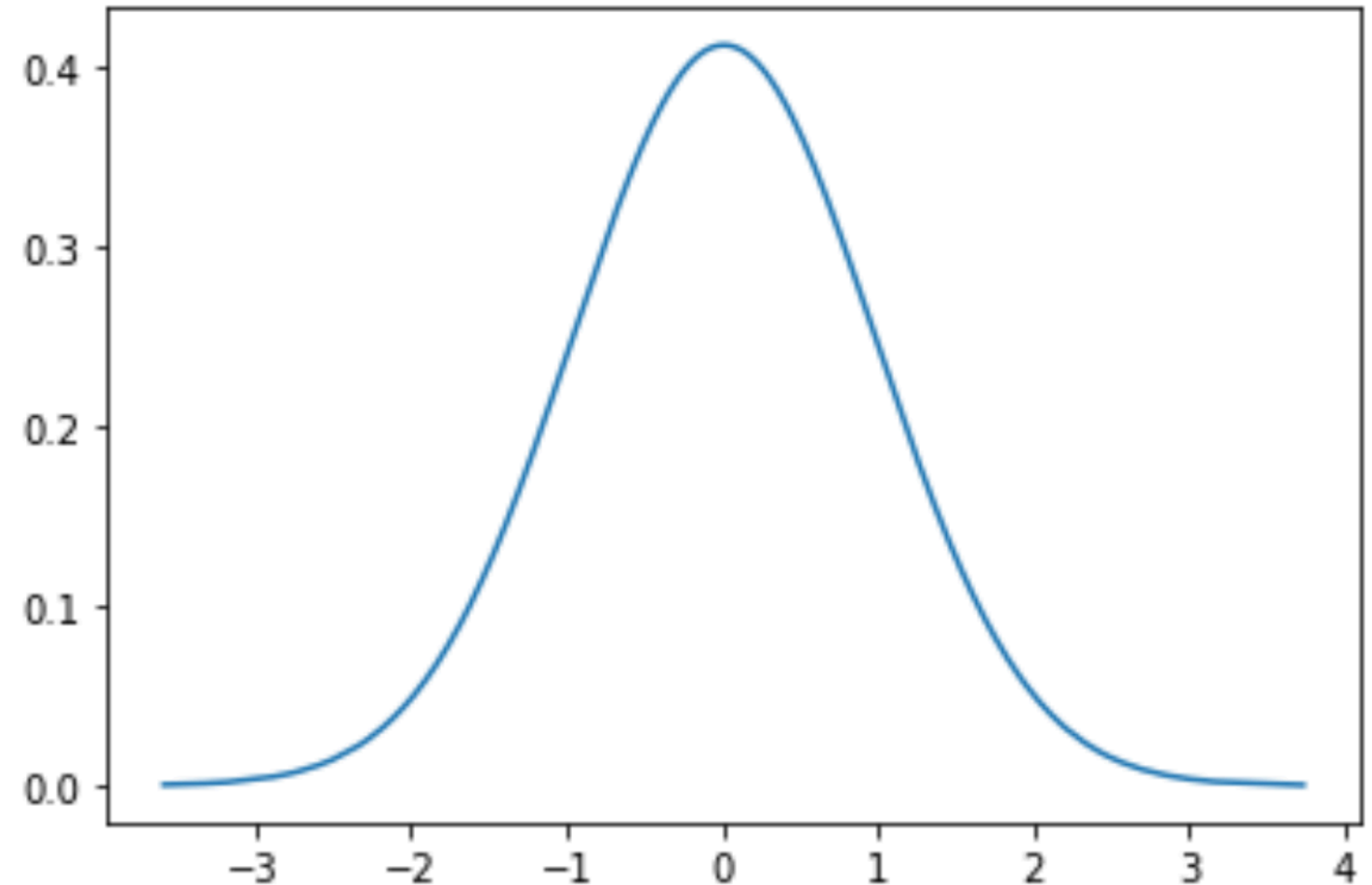
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The normal distribution

- The familiar 'bell curve' shape
- Related to z-score work

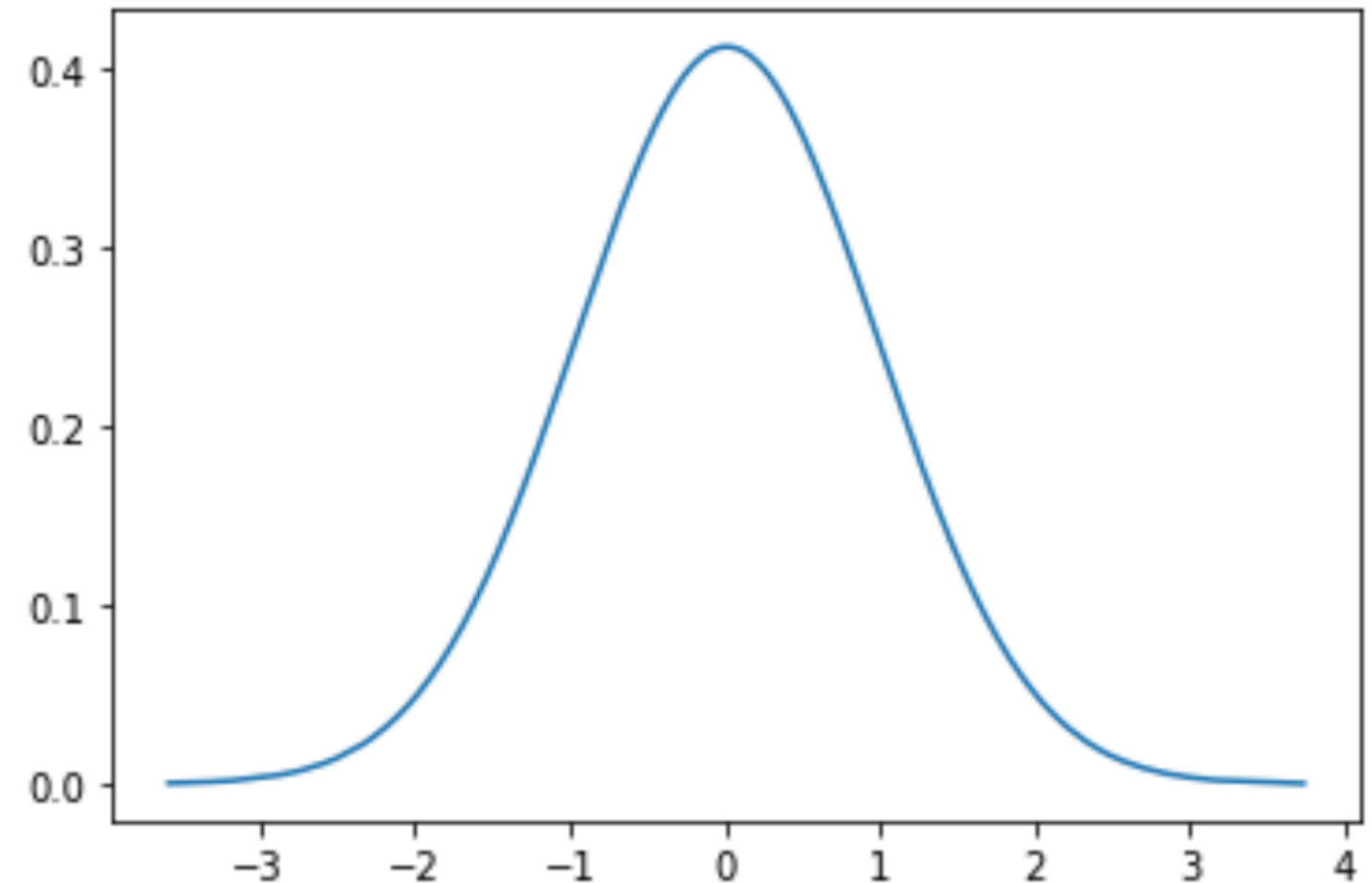
$$z = \frac{x - \mu}{\sigma}$$

- Mean = 0, std = 1
 - 'How many standard deviations is this point from the mean?'
 - 'What is the probability of obtaining this score?'



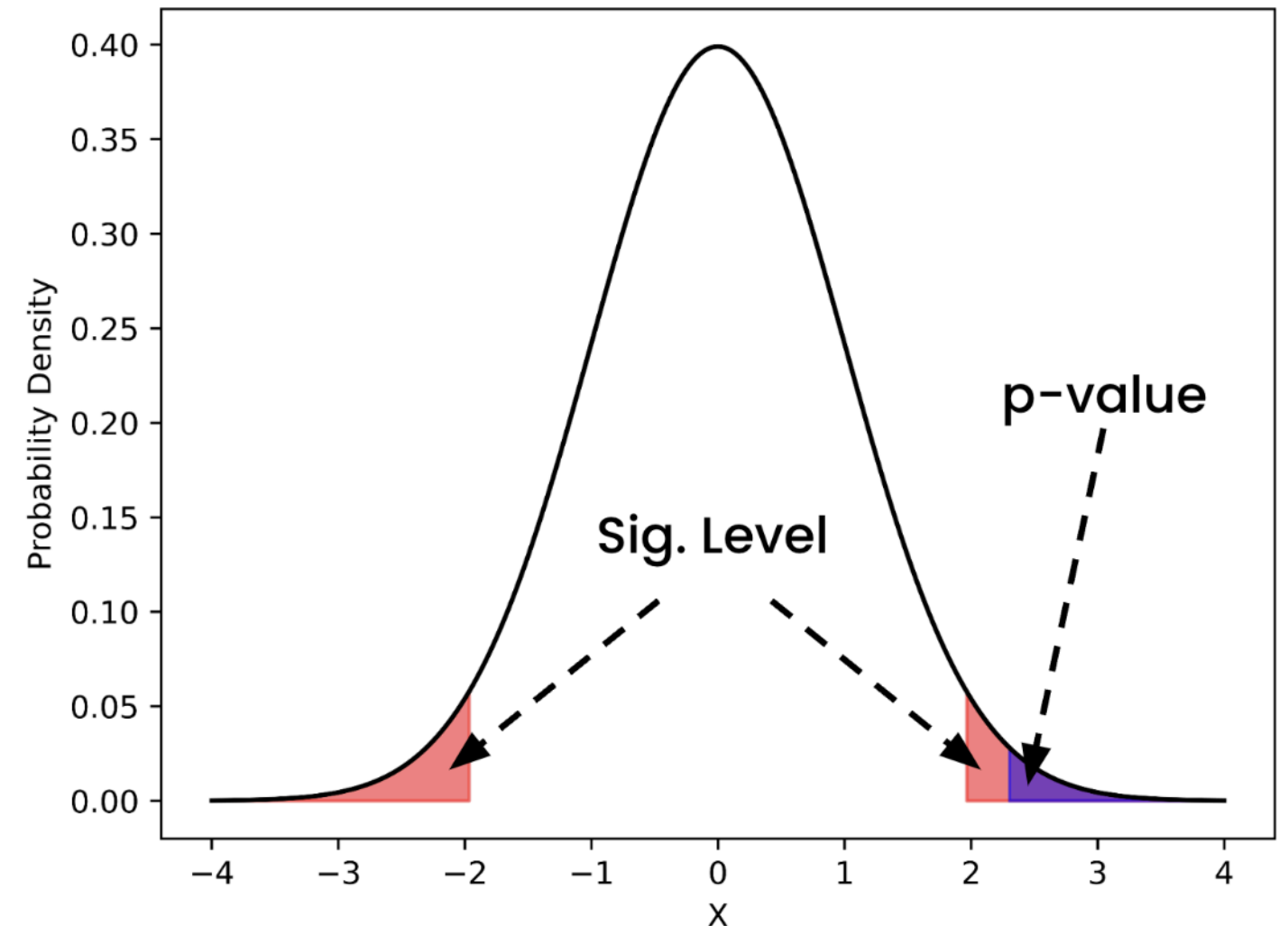
Normal data and statistical tests

- Required for **parametric** tests
- **Nonparametric tests:** don't assume normal data



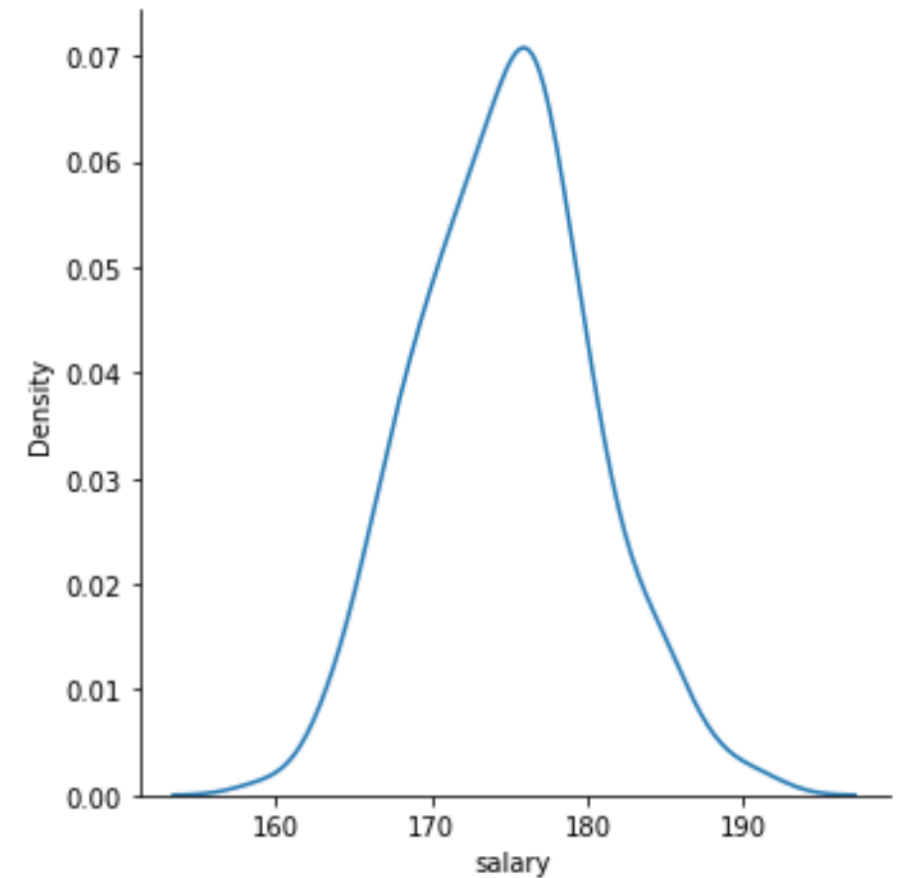
Normal, Z, and alpha

- Crucial link to significance level (α)
- Compare p-value to α
- Probability of a Type I error



Visualizing normal data

```
sns.displot(data=salaries,  
            x='salary',  
            kind="kde")  
  
plt.show()
```

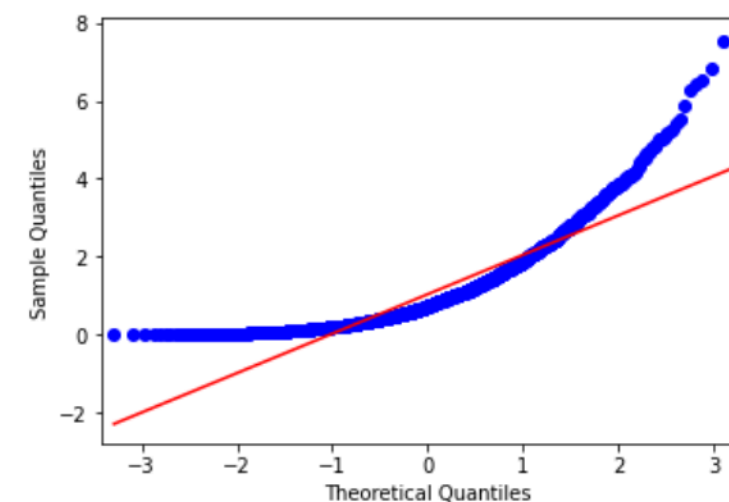
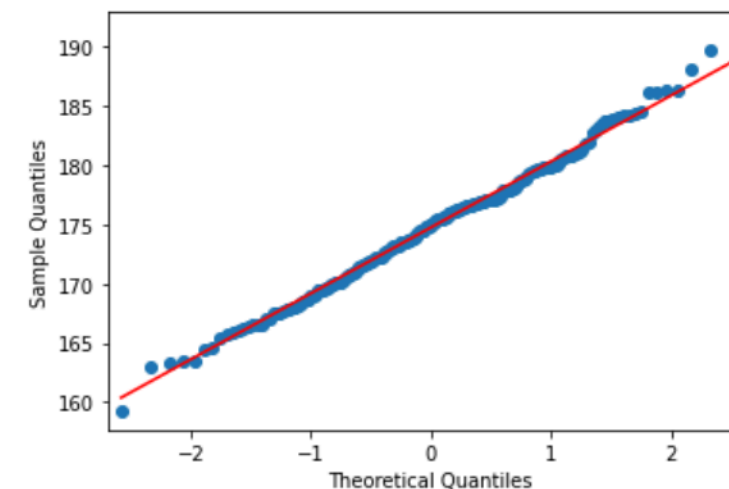


QQ plots

QQ plot: compare data to a particular distribution

```
from statsmodels.graphics.gofplots import qqplot
from scipy.stats.distributions import norm
qqplot(salaries['salary'],
       line='s',
       dist=norm)
plt.show()
```

- Ideal: dots hugging line
- Bad: bow out at ends



Tests for normality

- Shapiro-Wilk (good for smaller datasets)
- D'Agostino K^2 (uses **kurtosis** and **skewness**)
- Anderson-Darling (returns list of values)

H_0 = "Data is drawn from a Normal Distribution"

A Shapiro-Wilk test

```
from scipy.stats import shapiro
alpha = 0.05
stat, p = shapiro(salaries['salary'])
print(f"p: {round(p,4)} test stat: {round(stat,4)}")
```

```
p: 0.8293 test stat: 0.9956
```

- `p > alpha`
 - Fail to reject $H_0 \rightarrow$ likely normal

An Anderson-Darling test

```
from scipy.stats import anderson  
result = anderson(x=salaries['salary'], dist="norm")
```

```
print(round(result.statistic,4))  
print(result.significance_level)  
print(result.critical_values)
```

```
0.2748  
[15.  10.   5.   2.5  1. ]  
[0.572 0.651 0.781 0.911 1.084]
```

- `0.2748 < [0.572 0.651 0.781 0.911 1.084]`
 - Fail to reject $H_0 \rightarrow$ likely normal

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